

richbeam

MetaBeam32

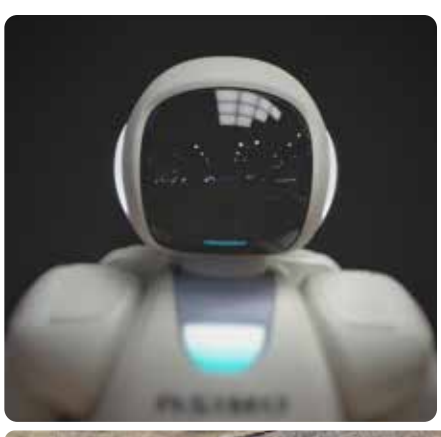
32-line Integrated LiDAR With
High Stability

Introduction

MetaBeam32 is a 32-line integrated LiDAR developed by Richbeam for Autonomous Driving and Robot, which is equipped with the LiDAR integrated receiving and emitting optical chip module independently developed by Richbeam, with stable performance and a very cost-effective price.

Applications

MetaBeam32 is suitable for indoor and outdoor commercial robot scenes such as cleaning, inspection, security, etc. It can be applied to scenes of commercial vehicles such as unmanned minibuses, logistics delivery, mining trucks, etc., and can also be used as a blind spot or main LiDAR for autonomous driving.







Core Parameters

Chip Sensor			
Scanning Principle	Mechanical Rotation	Relative Accuracy	2cm
Lines	32-line	Vertical FOV	31°(-16°~+15°)
Horizontal FOV	360°	Vertical Resolution	1°
Range	80m(10%) 120m(80%)	Range Accuracy	±5cm(0.05-1m) ±2cm(1-120m)
Returns	Single Return	Rotation Rate	10Hz，20Hz
Horizontal Resolution	0.24° 10Hz 0.48° 20Hz		

Output			
Data Transmission	UDP/IP Ethernet (100 Mbps)	Data Output	UDP: azimuth angle, distance, intensity, synchronized time stamp
Time Synchronization	$GPRMC with 1 PPS	PTP Clock Accuracy	⩽1us
Date Points	Single Return: 480,000points/sec	PTP Clock Drift	⩽1us/s
Clock Source	GPS/PTP		

Physical Parameters			
Laser Class	Class 1 Eye Safe	Dimension	H65mm*ϕ106mm
Wavelength	905nm	Environmental Protection	IP67
Operating Temperature	-25°C~+60°C	Operating Voltage	12~24V
Storage Temperature	-40°C~85°C	Power Consumption	8W
Weight	700g		

Integrated LiDAR Feature Highlights

01 Lower Price

Use the Mask Aligner to integrate dozens of optical chip channels to reduce the labor cost of optical adjustment; no labor loss is involved, reducing the trial and error costs; not paying for the redundant functions of ready-made modules, and reducing the cost of intermediaries.

02 Superior Performance

The integrated laser receiving and emitting chip module integrates multiple channels at the chip level to improve the product overall performance, with low power consumption, low system burden and long lifetime.

03 Small Volume

The product is highly integrated, the volume only accounts for 1/3 to 1/2 of the similar products in the general market, with light weight and easy to install on the front.

04 High Productivity

Instead of the manual debugging and components calibration, standardized production, the production speed is increased by at least 91%; the product consistency and stability are much higher than manual, and the angle error between the product channels is reduced by about 66%.

About Richbeam

Richbeam(Beijing) Technology Co., Ltd., formerly known as Yinbing Technology, was founded in 2016 and is an integrated chip LiDAR solutions provider. The founding team developed the world’s first integrated LiDAR based on nearly 20 years of scientific research experience in top universities. From the core technology, we continue to innovate and develop a variety of products. The products can be widely used in Autonomous Driving, Smart Transportation, Robotics, Industrial Automation, Smart Home and other fields. Richbeam has a complete intellectual property certification system, and now has more than 30 patents.

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